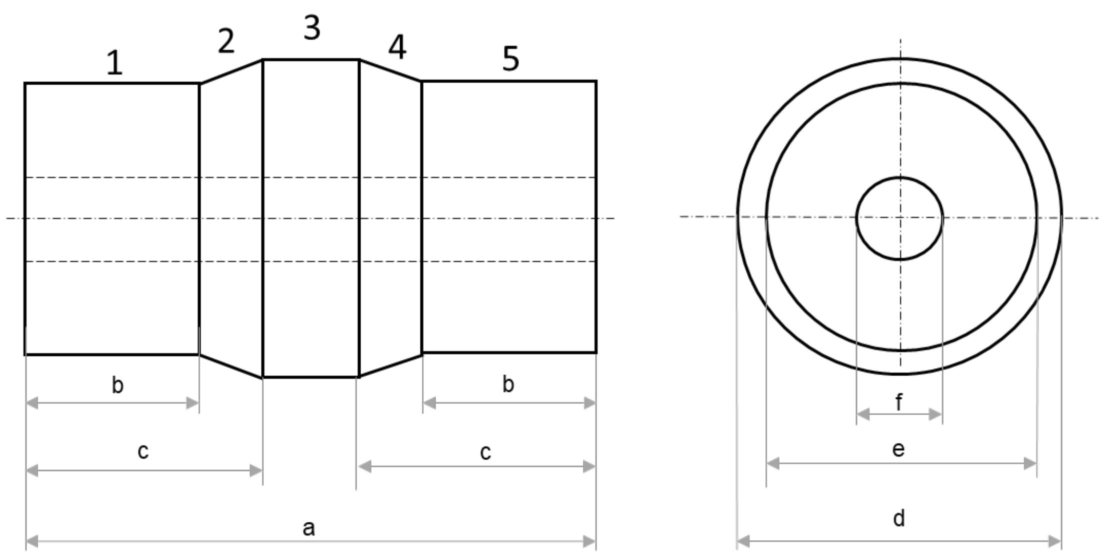
Problem 1 - 06/11/2020

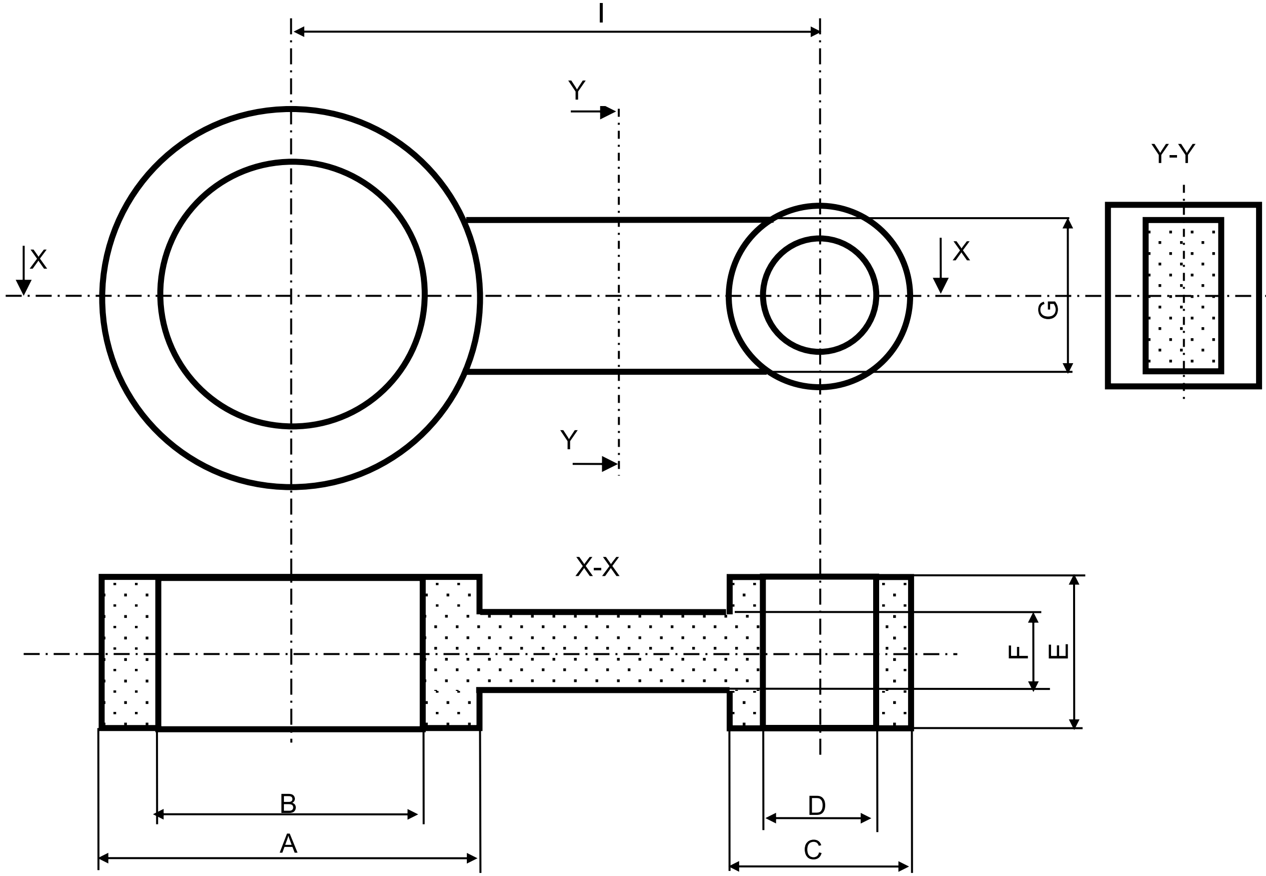
The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of AISI 304 steel (for simplicity, draft angles and fillets/radii are omitted). The casting is divided in 5 elementary geometries (numbered from 1 to 5).

1) Please, study the directional solidification of the metal. Dimensional data: a = 340 mm, b = 115 mm, c = 145 mm, d = 165 mm, e = 145 mm, f = 60 mm.
2) Considering: the casting axis as parting line, A_s:A_r:A_g = 2:2:1, c = 0,6, two gates with an area equal to 700 mm^2 each, a volume of the casting equal to 5 dm^3, a top riser with a volume equal to 3 dm^3, a cope height equal to 300 mm. Design a middle gating system, verifying if the mould filling time is lower than 10 s.



Problem 2 - 15/01/2021 (1)

The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of steel and will have the following final dimensions after machining: A = 50 mm, B = 35 mm, C = 24 mm, D = 15 mm, E = 20 mm, F = 10 mm, G = 20 mm, I = 70 mm.

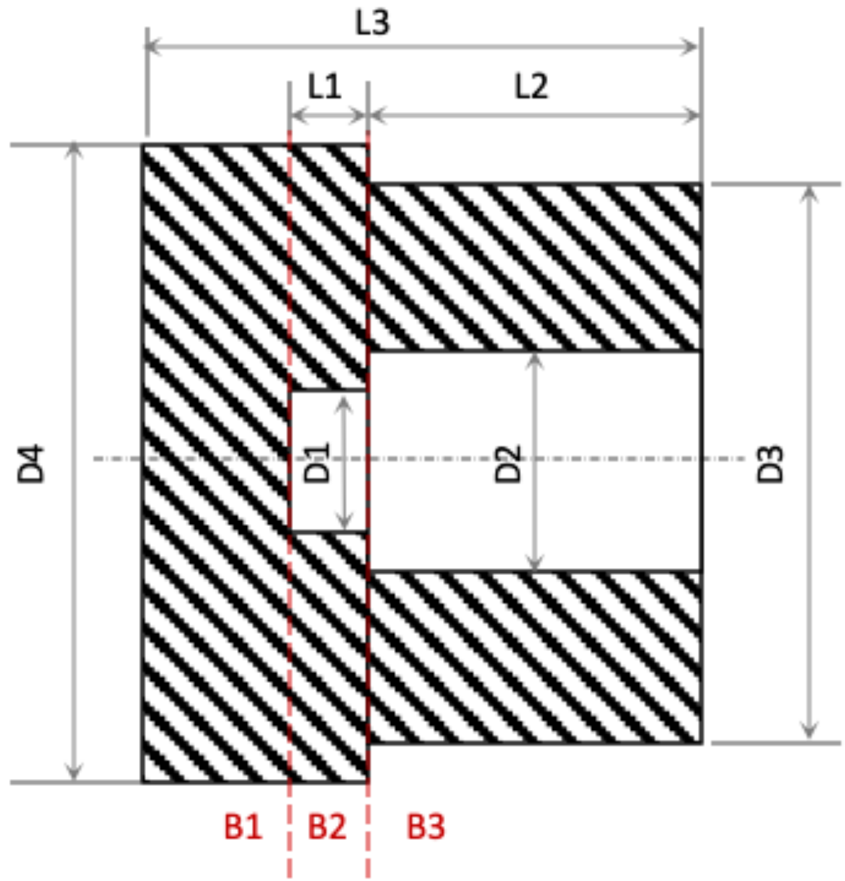


Problem 2 - 15/01/2021 (2)

1) Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).
2) Considering the original dimensions, study the directional solidification of the metal. Is the directional solidification feasible? (Consider three elements: Element A is the cylinder of external diameter A; Element C is cylinder of external diameter C, Element B is the middle block to be considered tangent to the cylinders)
3) Considering: A_s:A_r:A_g = 4:3:2, c = 0,5, a volume of the casting equal to 36 cm^3, a top riser with a volume equal to 15 cm^3, a distance of the upper plane of the mould from the lower plane of the cavity equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

Problem 3 - 05/02/2021 (1)

The figure shows an axisymmetric casting to be fabricated by an expendable mould process.
Data: $D1$ = 50 mm, $D1$ = 80 mm, $D3$ = 200 mm, $D4$ = 230 mm, $L1$ = 30 mm, $L2$ = 120 mm, $L3$ = 200 mm.



Problem 3 - 05/02/2021 (2)

1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).
2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?
3) Considering: the axis of the casting as parting line, a volume of the casting cavity equal to 6,5 dm^3, a riser with a volume equal to 3 dm^3, a middle gating system with H_{av} equal to 100 mm and c = 0,6. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.

Problem 4 - 12/07/2021

A steel lift pulley is obtained by sand casting using a middle gating system.
The top view and the cross-section of the main cavity in the mould are shown in the figure, where *PdS* is indicating the parting line.
In the figure the decomposition into elementary geometries is also indicated (draft angles and fillet radii are neglected).

1) Study the directional solidification of the casting.
2) Considering a volume of the entire cavity equal to $1{,}25 \cdot 10^6$ mm^3, a total volume of open risers equal to 400 cm^3, an initial pouring height equal to 100 mm, a friction factor c = 0,6, and a gating system ratio A_s:A_r:A_g = 4:3:2, design the gating system to ensure a mould filling time equal to 10 s.